

The Price of Proximity

Subway Access and NYC Apartment Rents

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Research Question

Among New York City rental listings in May 2026, what is the descriptive relationship between advertised monthly rent and straight-line distance to the nearest subway station, and how does that relationship change when apartment size and borough are included?



Data Sources and Preparation

Rental Data



- FirstMover NYC
- 20,385 listings
- Rent & apartment characteristics
- Coordinates

Transit Data



- Official MTA stations
- Station coordinates
- Nearest-station matching

Data Processing



- Merged datasets
- Calculated nearest subway distance
- Removed duplicates
- Cleaned missing values
- Train/test split

Final Dataset

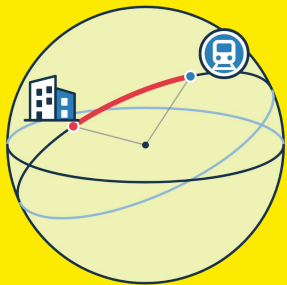


- Apartment features
- Subway distance
- Regression-ready dataset

Calculating Nearest Subway Distance

We calculated each rental's straight-line distance to the nearest subway station using the **Haversine Formula**

1. Compare Coordinates
 - Use the latitude and longitude of each rental and station
2. Calculate Distance



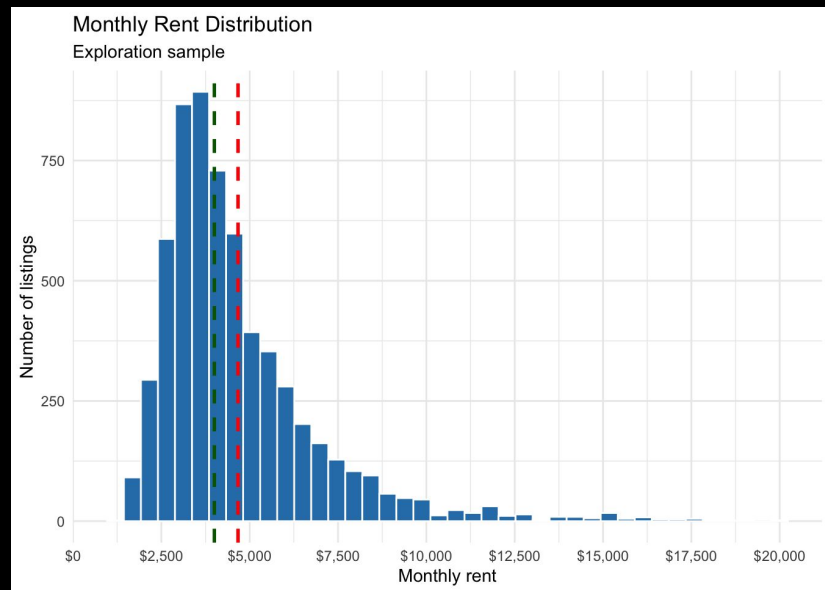
$$a = \sin^2(\Delta\phi/2) + \cos(\phi_1) \cos(\phi_2) \sin^2(\Delta\lambda/2)$$
$$d = 2R \operatorname{atan2}(\sqrt{a}, \sqrt{1-a})$$

3. Select the Nearest Station
 - Identify the station with the smallest distance to the rental

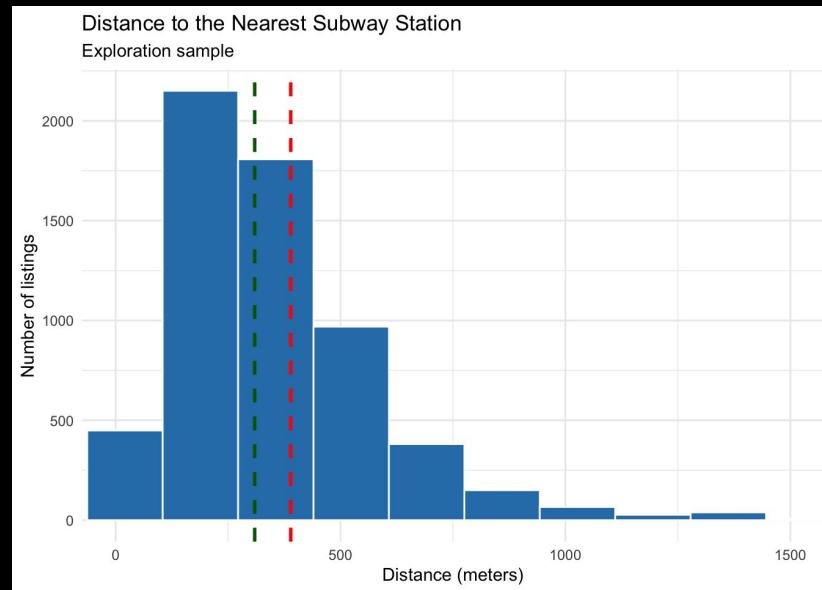


Variable Distributions

Monthly Rent



Distance to Nearest Subway



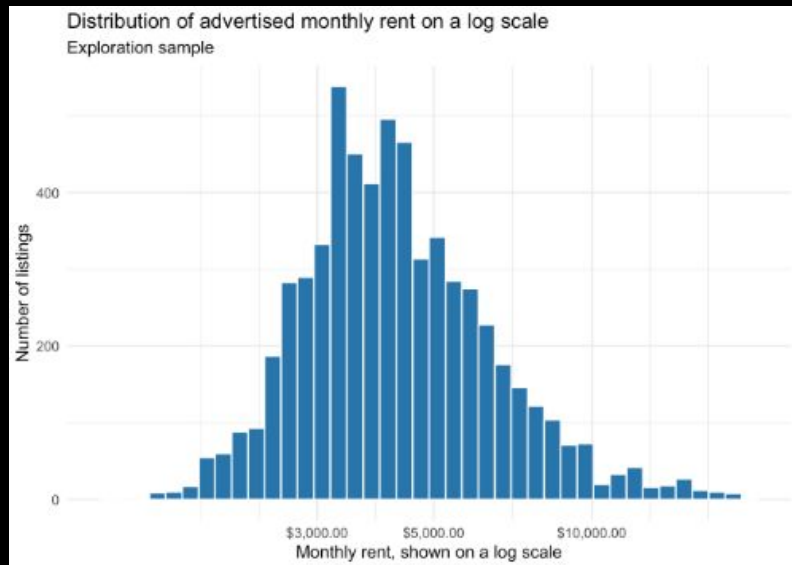
Green Dashed Line = Median



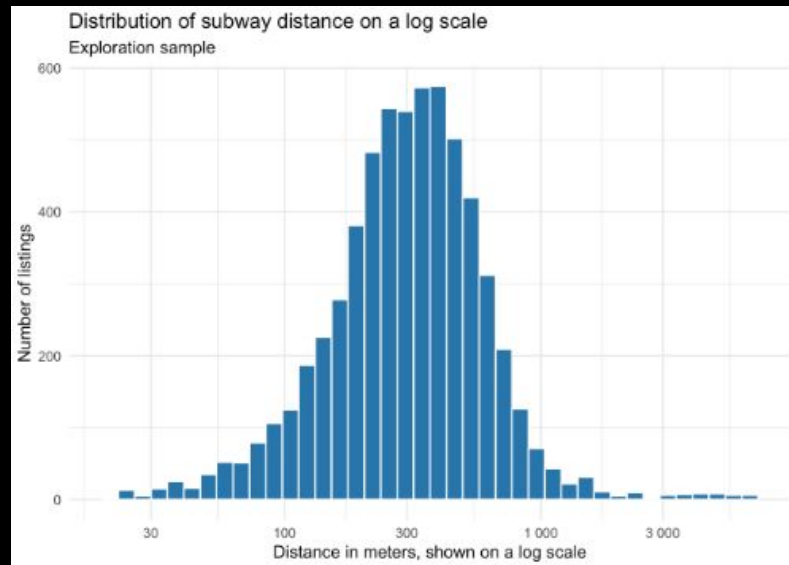
Red Dashed Line = Mean

Transformed Variable Distributions

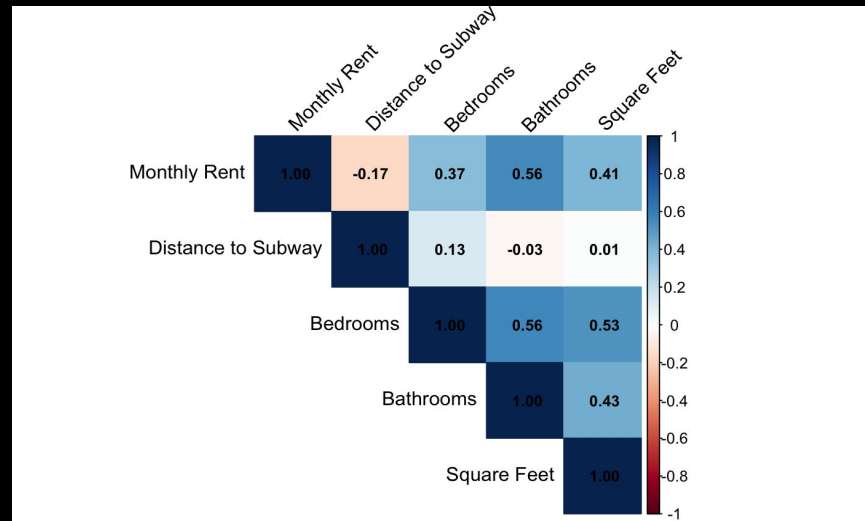
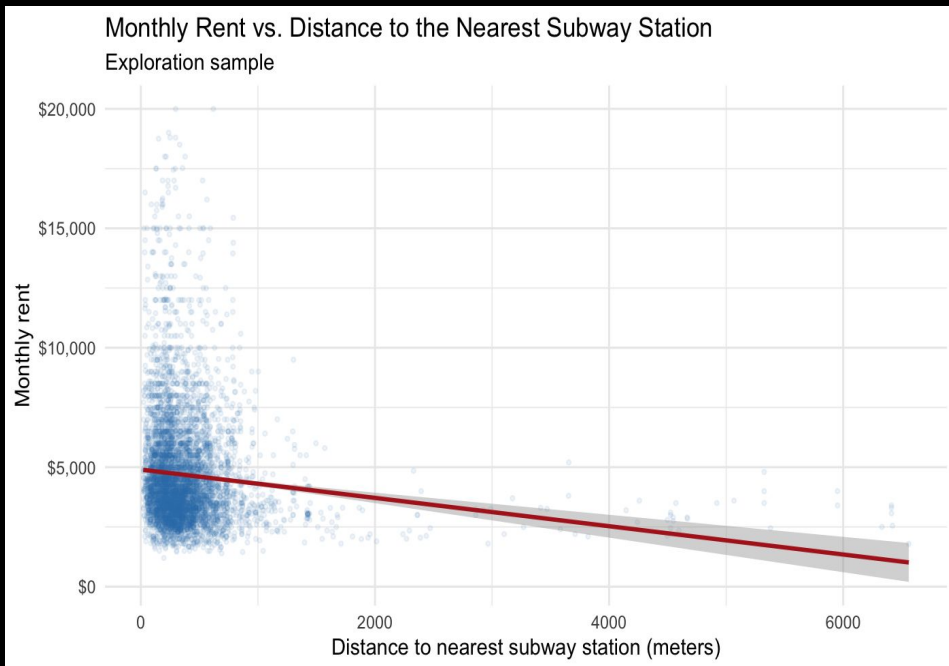
Log(Monthly Rent)



Log(Distance to Nearest Subway)



Relationship Among Key Variables



- Weak negative relationship between rent and subway distance
- Bathrooms show the strongest positive association with rent
- Low correlations among predictors suggest limited multicollinearity

From Raw Association to Final Model

Baseline Relationship

$$\text{Rent}_i = \beta_0 + \beta_1 \text{Distance}_i + \varepsilon_i$$

Measures the average rent difference associated with each additional meter from the nearest subway station

Specification Testing

Model Selection

- Linear model
- Log-distance model
- Log-rent model
- Log-log model

Final choice: Log-log specification

Final Adjusted Model

$$\log(\text{Rent}_i) = \beta_0 + \beta_1 \log(\text{Distance}_i) + \beta_2 \text{Bathrooms}_i + \beta_3 \text{Bedrooms}_i + \beta_4 \text{Sqft}_i + \beta_5 \text{Brooklyn}_i + \beta_6 \text{Manhattan}_i + \beta_7 \text{Queens}_i + \beta_8 \text{StatenIsland}_i + \varepsilon_i$$

Estimates the distance elasticity after accounting for apartment size and broad geography.

Final estimation: 3,101 confirmation listings · HC1 robust standard errors · descriptive, not causal

Key Findings

Subway Proximity Matters

A 10% increase in distance to the nearest subway station is associated with approximately a **0.71%** lower advertised monthly rent.

Apartment Characteristics Matter

+1 Bedroom → ≈4% higher rent

+1 Bathroom → ≈36% higher rent

+100 Square Feet → ≈3% higher rent

} High correlation between these variables – multicollinearity makes it difficult to isolate the coefficients

Compared with otherwise similar apartments in the Bronx:

Brooklyn → ≈35% higher rent

Manhattan → ≈89% higher rent

Queens → ≈11% higher rent

Staten Island → ≈27% lower rent

Model Performance

$R^2 = 0.616$

Robust standard errors

Association is **descriptive, not causal**

Final regression results	
=====	
	Dependent variable:

	Log monthly rent

Log distance to subway	-0.071*** (0.006)
Bedrooms	0.040** (0.019)
Bathrooms	0.308*** (0.032)
Square feet	0.0003*** (0.0001)
Brooklyn	0.300*** (0.035)
Manhattan	0.637*** (0.035)
Queens	0.102*** (0.035)
Staten Island	-0.313*** (0.069)
Constant	7.747*** (0.061)

Observations	3,101
R2	0.616
Adjusted R2	0.615
=====	
Note:	*p<0.1; **p<0.05; ***p<0.01

Assumptions and Limitations

Model Assumptions

- Linear relationship after log transformation
- Independence of rental listings
- HC1 robust standard errors used to address heteroskedasticity
- **Final specification selected after comparing alternative models**

Study Limitations

- Advertised listing prices, not final lease prices
- Straight-line distance rather than walking distance or travel time
- Other neighborhood factors (schools, amenities, safety) were not included
- Results describe associations, **not causal effects**

These findings should be interpreted as descriptive evidence of market patterns rather than causal estimates of subway access.

Conclusions

Apartments that are closer to subway stations generally have higher advertised rents, even after accounting for apartment size and borough.

- ✓ Subway proximity is associated with higher advertised rents.
- ✓ Apartment size and borough explain a much larger share of rent variation.
- ✓ The relationship is **descriptive**, not **causal**.



Thank You!